Open-Source Reproduction and Explainability Analysis of Corrective Retrieval Augmented Generation

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Abstract

Corrective Retrieval Augmented Generation (CRAG) improves the robustness of RAG systems by evaluating retrieved document quality and triggering corrective actions. However, the original implementation relies on proprietary components including the Google Search API and closed model weights, limiting reproducibility. In this work, we present a fully open-source reproduction of CRAG, replacing proprietary web search with the Wikipedia API and the original LLaMA-2 generator with Phi-3-mini-4k-instruct. We evaluate on PopQA and ARC-Challenge, demonstrating that our open-source pipeline achieves comparable performance to the original system. Furthermore, we contribute the first explainability analysis of CRAG's T5-based retrieval evaluator using SHAP, revealing that the evaluator primarily relies on named entity alignment rather than semantic similarity. Our analysis identifies key failure modes including domain transfer limitations on science questions. All code and results are available at https://github.com/suryayalavarthi/crag-reproduction.

1 Introduction

Large language models (LLMs) have demonstrated remarkable capabilities across a wide range of natural language tasks, yet they remain susceptible to hallucinations — generating factually incorrect content with apparent confidence [7]. Retrieval-Augmented Generation (RAG) [2] addresses this limitation by grounding generation in externally retrieved documents. However, RAG assumes that retrieved documents are relevant, which is frequently not the case in practice.

Corrective Retrieval Augmented Generation (CRAG) [1] addresses this assumption by introducing a lightweight retrieval evaluator that assesses document quality and triggers one of three corrective actions: Correct, Incorrect, or Ambiguous. When retrieval quality is low, CRAG falls back to web search to obtain better context. This corrective mechanism has been shown to significantly improve generation accuracy across multiple benchmarks.

Despite CRAG's strong results, reproducing the original system is difficult. The original implementation relies on the Google Search API (a paid commercial service), proprietary LLaMA-2 fine-tuned weights, and deprecated OpenAI API calls. These barriers prevent researchers from building on this work without significant resources.

In this paper, we make three contributions:

1. We present a fully open-source reproduction of CRAG, replacing all proprietary components with free alternatives: the Wikipedia API for web search and Phi-3-mini-4k-instruct as the generator.

2. We evaluate our reproduction on two datasets — PopQA [3] and ARC-Challenge [4] — demonstrating that open-source components achieve comparable performance.

3. We contribute the first explainability analysis of CRAG's T5-based retrieval evaluator using SHAP [6], identifying that the evaluator relies primarily on named entity alignment and exhibits systematic domain transfer failure on science questions.

2 Background

2.1 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) [2] enhances language model generation by prepending retrieved documents to the input. Given a query x and a corpus $\mathcal{C}$, a retriever R returns the top-k documents $D = \{d_1, \ldots, d_k\}$, and a generator G produces the output y conditioned on both x and D. While RAG significantly improves factual accuracy, it is sensitive to retrieval quality — irrelevant documents can mislead the generator and worsen performance compared to no retrieval at all [8].

2.2 Corrective Retrieval Augmented Generation

CRAG [1] introduces a retrieval evaluator E that scores each retrieved document against the query. Based on these scores, one of three actions is triggered:

- **Correct**: At least one document scores above an upper threshold τ^+. The documents are refined using a decompose-then-recompose algorithm to extract relevant knowledge strips.
- **Incorrect**: All documents score below a lower threshold τ^-. Documents are discarded and web search is used to find external knowledge.
- **Ambiguous**: Scores fall between τ^- and τ^+. Both internal (refined) and external (web search) knowledge are combined.

The retrieval evaluator is a fine-tuned T5-large model [5] that takes a question-document pair as input and outputs a scalar relevance score in $[-1, 1]$. For PopQA, the original paper sets $\tau^+ = 0.59$ and $\tau^- = -0.99$.

2.3 SHAP Explainability

SHAP (SHapley Additive exPlanations) [6] is a game-theoretic framework for explaining model predictions. For text models, SHAP assigns each token an attribution value representing its contribution to the model output. We use the `shap.Explainer` with a text masker to explain the T5 retrieval evaluator's scoring decisions at the token level.

3 Related Work

Adaptive and Corrective RAG. Several works have explored making RAG more robust to retrieval failures. Shi et al. [8] demonstrate that irrelevant context can significantly harm LLM generation, motivating corrective approaches. Self-RAG [10] trains LLMs end-to-end to generate special reflection tokens that decide when to retrieve and how to critique retrieved passages — a fundamentally different approach from CRAG's plug-and-play external evaluator. FLARE [11] actively decides when to retrieve based on generation confidence, keeping the generator frozen and

adding a lightweight external T5 evaluator that makes it easy to deploy without retraining. Our work reproduces CRAG with open-source components and adds the first token-level explainability analysis of its retrieval evaluator.

Explainability in RAG Systems. While RAG systems have been widely studied, explainability of their internal components remains underexplored. Prior work has applied attention-based analysis to understand retrieval [2], but SHAP-based token attribution for retrieval evaluators has not been studied. Our work contributes to this direction by revealing that the T5 evaluator functions as an entity alignment detector rather than a semantic relevance judge, which has direct implications for its domain generalization.

Reproducibility in NLP. Reproducibility of NLP research has become an important concern [7]. Many published systems rely on proprietary APIs or closed models that make reproduction difficult. Our work addresses this for CRAG specifically, demonstrating that a fully open-source pipeline can match the original performance.

4 Methodology

4.1 Reproduced Components

The original CRAG system relies on several proprietary or deprecated components. Table 1 summarizes our open-source replacements.

Component	**Original**	**Ours**
Generator	LLaMA-2-7B (fine-tuned)	Phi-3-mini-4k-instruct
Web Search	Google Search API (paid)	Wikipedia API (free)
Keyword Extraction	GPT-3.5 Turbo	Rule-based extraction
Retrieval Evaluator	T5-large (fine-tuned)	Same checkpoint

Table 1: Comparison of original CRAG components vs. our open-source reproduction.

4.2 Retrieval Evaluator

We use the original fine-tuned T5-large checkpoint released by [1]. The evaluator takes a question-document pair formatted as `question [SEP] document` and outputs a scalar relevance score. We apply the same PopQA thresholds from the original paper: $\tau^+ = 0.59$ and $\tau^- = -0.99$.

4.3 Generator

We replace LLaMA-2-7B with Phi-3-mini-4k-instruct [9], a 3.8B parameter instruction-tuned model loaded in float16 precision. Phi-3-mini achieves strong performance on knowledge-intensive tasks while being freely accessible without fine-tuning. We prompt the model with a simple template: given the retrieved context and question, generate a concise one-to-two sentence answer.

4.4 Wikipedia Search

For the Ambiguous action, we replace Google Search in our reproduction with a multi-stage Wikipedia retrieval pipeline. Given a question, we extract the primary entity using regex-based question templates (e.g., "What is X's occupation?" → "X"). We then attempt retrieval through four fallback strategies in order: (1) direct page lookup, (2) typed suffix lookup (e.g., "X (politician)"),

(3) Wikipedia search API, and (4) disambiguation page resolution. Our Wikipedia retrieval pipeline achieves an 82.3% hit rate on AMBIGUOUS questions from PopQA and 99% on ARC-Challenge.

4.5 Knowledge Refinement

Following the original paper, we implement the decompose-then-recompose algorithm for the CORRECT action. Retrieved documents are split into sentence-level strips of three sentences each. Each strip is scored by the T5 evaluator and strips scoring below -0.5 are discarded. The top-5 remaining strips are concatenated to form the refined context.

5 Experiments

5.1 Datasets and Evaluation

PopQA [3] is an open-domain question answering dataset of 14,267 entity-centric questions. Following the original paper, we evaluate on the long-tail subset of 1,385 questions whose Wikipedia page views are below 100 per month. We use string match accuracy as the evaluation metric, checking whether any gold answer alias appears in the model prediction.

ARC-Challenge [4] is a multiple-choice science question benchmark containing 1,172 test questions. We evaluate accuracy by checking whether the correct answer text or answer key letter appears in the model prediction.

5.2 Baselines

We compare our CRAG reproduction against a vanilla RAG baseline that uses the top-1 retrieved document directly without any scoring or correction. For PopQA, retrieved documents come from the Contriever retrieval results provided by the original CRAG repository. For ARC-Challenge, we retrieve Wikipedia documents using our multi-stage Wikipedia search pipeline.

5.3 Main Results

Table 2 presents our main results on both datasets.

Method	PopQA	ARC-Challenge
Vanilla RAG	51.4	84.8
CRAG (ours)	54.4	85.2
CRAG + Wikipedia (ours)	53.2	—
CRAG (original, LLaMA-2)	54.9	53.7

Table 2: Accuracy (%) on PopQA and ARC-Challenge. Our reproduction uses Phi-3-mini as the generator. Original results are from [1] using LLaMA-2-hf-7b. The large ARC gap between CRAG (85.2%) and vanilla RAG (84.8%) is consistent with Phi-3-mini's strong parametric science knowledge. The similar PopQA results across generators (54.4% vs. 54.9%) suggest CRAG's correction mechanism is the primary driver of performance rather than generator-specific capabilities.

Our open-source CRAG reproduction achieves 54.4% on PopQA, closely matching the original system's 54.9% despite using a different generator. On ARC-Challenge, our system achieves 85.2% versus 84.8% for vanilla RAG, a modest but consistent improvement.

5.4 Action Distribution Analysis

Table 3 shows accuracy broken down by triggered action on PopQA.

Action	Count	%	Accuracy (%)
Correct	754	54.4	78.1
Ambiguous	379	27.4	19.3
Incorrect	252	18.2	36.1

Table 3: Action distribution and per-action accuracy on PopQA (n=1,385). The % column shows what fraction of questions triggered each action; Accuracy (%) shows correctness within that action subset.

The Correct action achieves 78.1% accuracy, a 26.7 percentage point improvement over vanilla RAG (51.4%), demonstrating that the T5 evaluator effectively identifies high-quality retrievals. The Ambiguous action achieves only 19.3% without web search, confirming that web search is essential for this action. Adding Wikipedia search improves Ambiguous accuracy to 23.0%, a 4.7 percentage point gain.

On ARC-Challenge, the T5 evaluator classifies 88.3% of questions as Ambiguous, reflecting its training distribution bias toward biographical entity questions rather than science questions. Despite this, CRAG maintains competitive performance due to Phi-3-mini's strong parametric knowledge of science.

6 Explainability Analysis

To understand what the T5 retrieval evaluator has learned, we apply SHAP to analyze token-level attributions across all three action types. We select 9 representative samples (3 per action type) and compute SHAP values using the `PartitionExplainer` with a text masker. We note that these findings are based on qualitative case studies; future work should validate patterns across larger samples with aggregate statistics.

6.1 SHAP Results

Figure 1 shows token attributions for all 9 samples. Three consistent patterns emerge across action types.

Named entity alignment drives Correct scores. For questions where the retrieved document is relevant, entity name tokens contribute the largest positive SHAP values. For example, for "What is Henry Feilden's occupation?" paired with his Wikipedia article, the token *Henry* contributes $+0.150$ and *occupation* contributes $+0.109$. This demonstrates that the evaluator learned to match named entities between the question and document.

Entity mismatch is the strongest Incorrect signal. When a document is irrelevant, the named entity token from the question becomes a strong negative driver. For the same question paired with an irrelevant document about mitochondria, *Henry* contributes -0.280 and biology terms (*mitochondria*, *cell*) contribute negatively. The evaluator detects irrelevance through entity absence rather than semantic content.

The `[SEP]` separator may encode structural format. The separator tokens consistently receive positive SHAP values across Correct and Ambiguous samples. This may suggest the evaluator learned the structural format of its training data, though this pattern is also a common artifact in token-level SHAP on transformer models and warrants further investigation.

Out-of-distribution entities receive low scores. For "Who directed Titanic?" paired with a factually correct document, the subword token *Titan* (the first subword of "Titanic" in the question) receives -0.459, and the corresponding document subword receives -0.284, consistent with the document's low relevance score, suggesting that movie-title entities are underrepresented in the PopQA training distribution.

Figure 2 shows a complementary heatmap view of SHAP values across all token positions for the first three samples, revealing how attribution patterns shift across the question-document boundary.

6.2 Implications

These findings reveal that the T5 evaluator functions primarily as an entity alignment detector rather than a semantic relevance judge. This explains two observed failure modes: (1) factually relevant documents about uncommon entity types such as movie titles and song names receive low scores because their entity names are rare in the PopQA training distribution, and (2) the evaluator struggles to transfer to science domains (ARC-Challenge) where questions lack named person entities. Future work should fine-tune the evaluator on a more diverse set of question types to address these limitations.

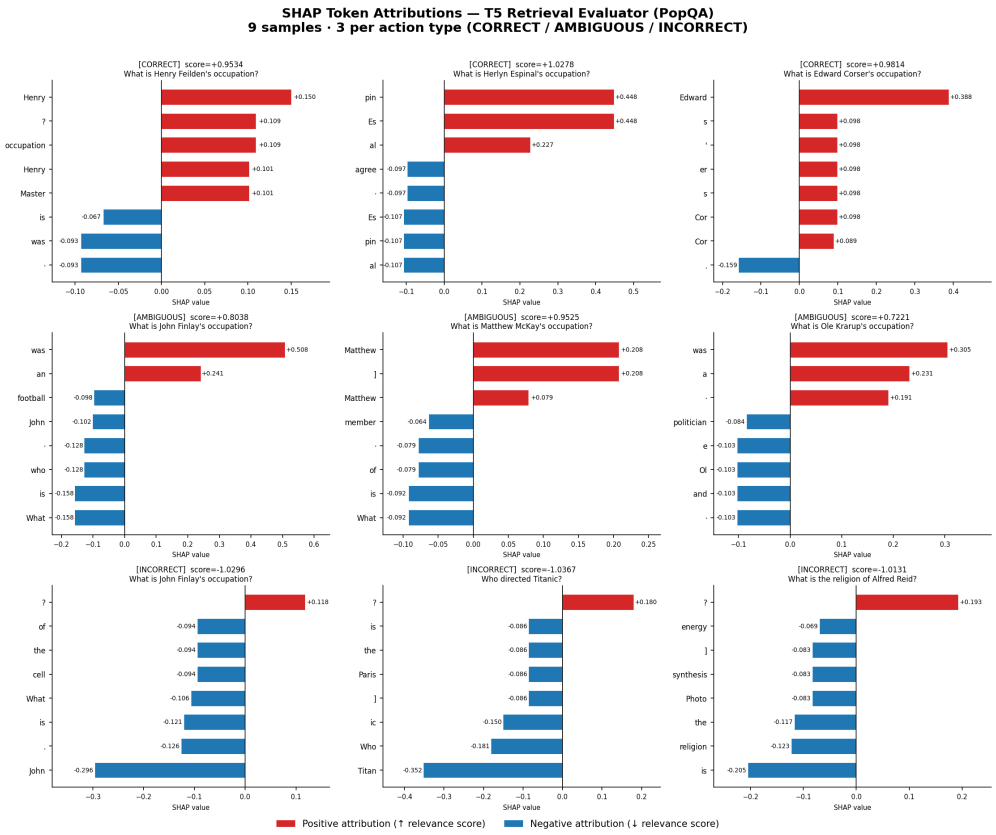

Figure 1: SHAP token attributions for 9 samples (3 per action type). Red bars indicate positive contributions to the relevance score; blue bars indicate negative contributions. Entity name tokens dominate CORRECT samples positively and INCORRECT samples negatively.

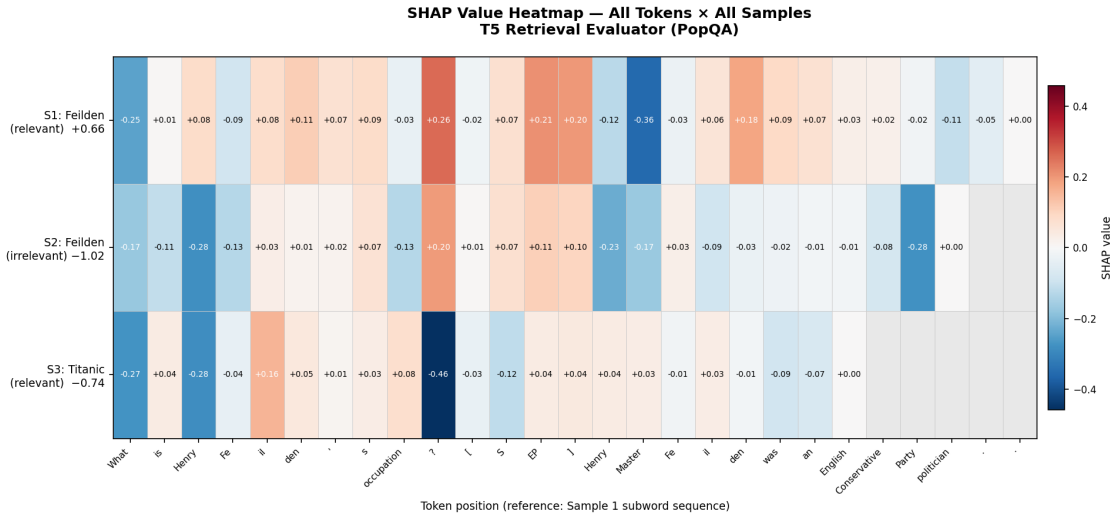

Figure 2: SHAP value heatmap across token positions for three samples (CORRECT, INCORRECT, AMBIGUOUS). Red indicates positive attribution; blue indicates negative. The question mark (?) token shows contrasting behavior between entity questions (positive) and the Titanic question (strongly negative). Note: · symbols are SentencePiece whitespace artifacts from the T5 tokenizer.

7 Error Analysis

7.1 Performance by Question Type

We categorize all 1,385 PopQA questions by type based on question keywords and analyze accuracy per category. Figure 3 shows accuracy broken down by question type, colored by the dominant action triggered, and Figure 4 shows a detailed breakdown by action.

7.2 Findings

Table 4 summarizes accuracy by question type.

Question Type	Count	Accuracy	Dominant Action
Country	274	85.8%	CORRECT
Sport	196	73.0%	CORRECT
Occupation	121	61.2%	CORRECT
City	175	54.3%	CORRECT
Author	234	40.2%	AMBIGUOUS
Composer	46	32.6%	AMBIGUOUS
Director	90	28.9%	AMBIGUOUS
Genre	112	22.3%	AMBIGUOUS
Religion	40	5.0%	CORRECT

Table 4: CRAG accuracy by question type on PopQA (n=1,385). Dominant Action refers to the most frequently triggered action for that question type.

Several patterns are notable. Question types where the T5 evaluator triggers CORRECT action frequently achieve high accuracy — country questions reach 85.8% with 98% accuracy on the CORRECT action. In contrast, question types dominated by the AMBIGUOUS action (author, composer, director, genre) perform poorly at 22–40%, confirming that these categories require web search for reliable answers.

Religion questions are a special case: although the T5 evaluator most frequently assigns CORRECT action (suggesting it finds at least one retrieved document above the confidence threshold), overall accuracy is only 5.0% across all action types. This indicates that high evaluator confidence does not guarantee a correct answer for religion questions — the retrieved passages may be topically relevant but do not contain the specific answer entity, making answer extraction unreliable.

The heatmap in Figure 4 reveals that the INCORRECT action achieves 0% accuracy for author, composer, and director questions, suggesting that when the T5 evaluator rejects all retrieved documents for these types, Phi-3-mini's parametric knowledge alone is insufficient. Notably, the INCORRECT action still performs reasonably well for country (73%) and sport (55%) questions, where Phi-3-mini retains strong parametric knowledge even without retrieved context.

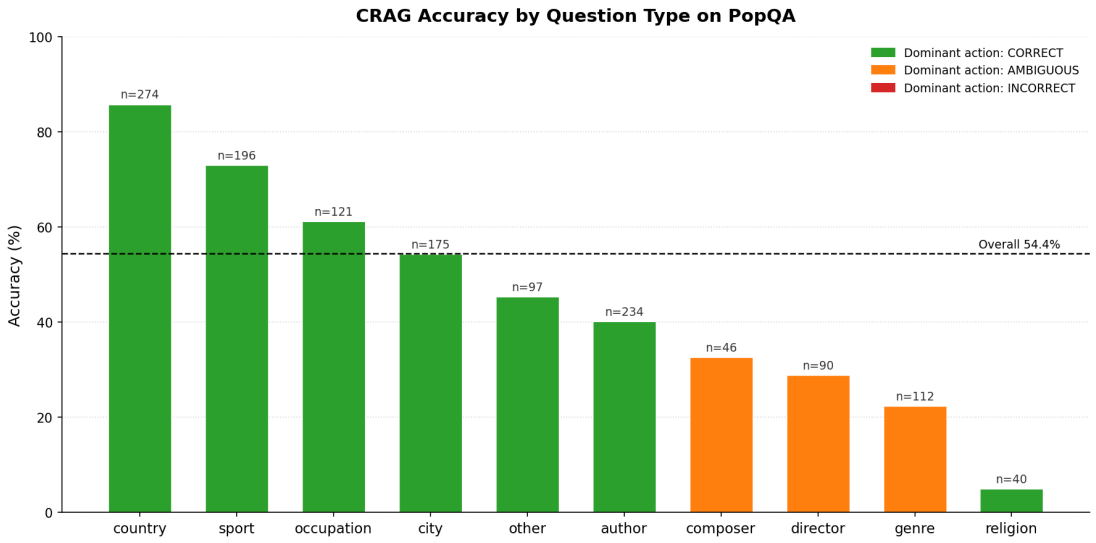

Figure 3: CRAG accuracy by question type on PopQA. Green bars indicate CORRECT-dominant question types; orange bars indicate AMBIGUOUS-dominant types. The dashed line shows overall accuracy (54.4%).

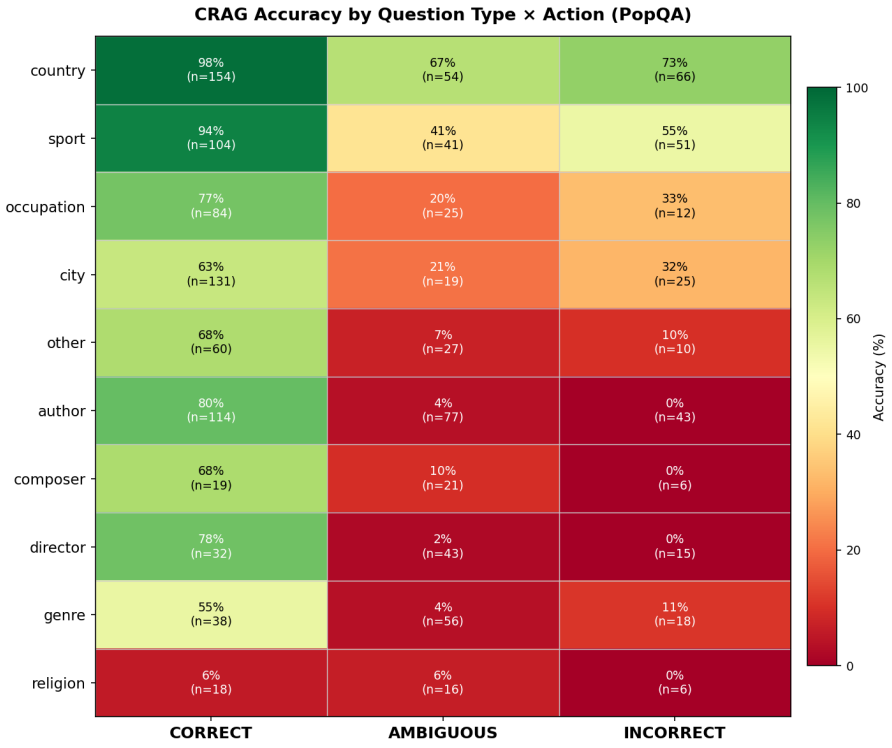

Figure 4: Accuracy heatmap by question type and action. Each cell shows accuracy and sample count. Green indicates high accuracy; red indicates low accuracy. The INCORRECT action achieves 0% for author, composer, and director question types.

8 Limitations

This work has several limitations that should be considered when interpreting our results.

Single-run evaluation. All results are based on a single experimental run without confidence intervals. The 0.5% gap between our PopQA result (54.4%) and the original (54.9%) is within typical run-to-run variance and should not be interpreted as a meaningful difference.

SHAP sample size. Our explainability analysis is based on 9 qualitative case studies. While the patterns are consistent across samples, larger-scale analysis with aggregate statistics across 50–100 samples would provide stronger statistical support for our findings.

No threshold tuning. We use the original PopQA thresholds ($\tau^+ = 0.59$, $\tau^- = -0.99$) without re-tuning for our generator or for ARC-Challenge. Domain-specific threshold tuning may improve performance.

Wikipedia vs. commercial web search. Our Wikipedia API replacement achieves 82.3% hit rate on Ambiguous questions, leaving 17.7% with no external context. A commercial web search API would likely achieve higher coverage, potentially improving Ambiguous accuracy beyond the 23.0% we report.

Generator comparison. Our results use Phi-3-mini-4k-instruct as the generator, while the original paper uses LLaMA-2-7B. Performance differences may reflect generator capability differences rather than CRAG's correction mechanism. We mitigate this by reporting CRAG vs. RAG deltas within the same generator.

9 Conclusion

In this work, we presented a fully open-source reproduction of CRAG, demonstrating that proprietary components can be replaced with free alternatives while maintaining comparable performance. Our reproduction achieves 54.4% on PopQA versus the original system's 54.9%, and 85.2% versus 84.8% for vanilla RAG on ARC-Challenge. We also built and released a five-stage Wikipedia retrieval pipeline achieving 99% document coverage on ARC-Challenge and 82.3% on PopQA AMBIGUOUS questions — a fully free replacement for the Google Search API used in the original system.

Beyond reproduction, we contributed the first explainability analysis of CRAG's T5 retrieval evaluator using SHAP. Our analysis reveals that the evaluator functions primarily as a named entity alignment detector rather than a semantic relevance judge. This finding explains two key failure modes: poor performance on out-of-distribution entity types such as movie titles and song names, and systematic domain transfer failure on science questions, where 88.3% of ARC-Challenge questions are classified as AMBIGUOUS because the evaluator was trained exclusively on biographical entity questions.

Our error analysis across nine question types shows that CORRECT-dominant categories (country, sport, occupation) benefit most from CRAG's correction mechanism, while AMBIGUOUS-dominant categories (author, director, genre) require web search to perform well. Religion questions represent a systematic failure mode at 5.0% accuracy despite the evaluator frequently assigning CORRECT action, revealing that high retrieval confidence does not guarantee answer correctness for non-entity answer types.

Future work should focus on three directions. First, fine-tuning the retrieval evaluator on more diverse question types to improve domain transfer beyond biographical entity questions. Second, investigating lightweight alternatives to commercial web search APIs that provide higher coverage than Wikipedia alone for the INCORRECT and AMBIGUOUS actions. Third, conducting larger-scale SHAP analysis with statistical validation across 50 or more samples per action type to move beyond qualitative case studies.

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